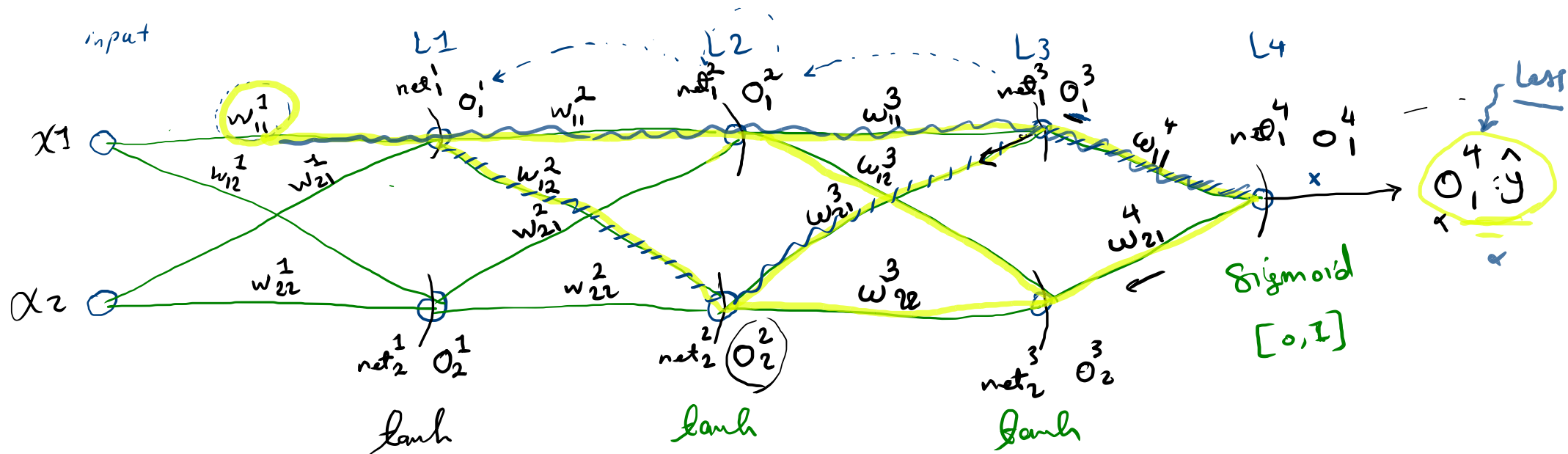


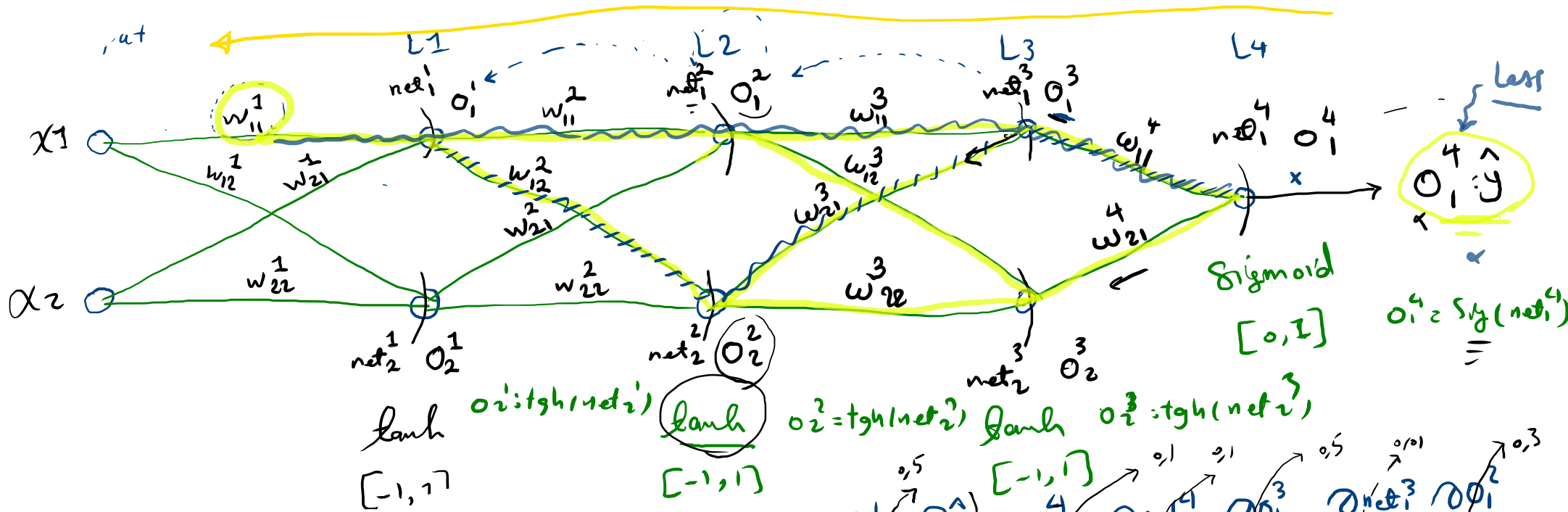


$$w_{11}^{1-new} = w_{11}^1 - \alpha \frac{\partial L}{\partial w_{11}^1}$$

path-1: $\frac{\partial L}{\partial w_{11}^1} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_1^4} \frac{\partial o_1^4}{\partial net_1^4} \frac{\partial net_1^4}{\partial o_1^3} \frac{\partial o_1^3}{\partial net_1^3} \frac{\partial net_1^3}{\partial o_1^2} \frac{\partial o_1^2}{\partial net_1^2} \frac{\partial net_1^2}{\partial o_1^1} \frac{\partial o_1^1}{\partial net_1^1} \frac{\partial net_1^1}{\partial w_{11}^1}$

path-2: $\frac{\partial L}{\partial w_{11}^1} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_1^4} \frac{\partial o_1^4}{\partial net_1^4} \frac{\partial net_1^4}{\partial o_1^3} \frac{\partial o_1^3}{\partial net_1^3} \frac{\partial net_1^3}{\partial o_2^2} \frac{\partial o_2^2}{\partial net_2^2} \frac{\partial net_2^2}{\partial o_1^1} \frac{\partial o_1^1}{\partial net_1^1} \frac{\partial net_1^1}{\partial w_{11}^1}$

$$\frac{\partial L}{\partial w_{ii}} = \left[\left(\frac{\partial L}{\partial w_{ii}} \right)_{\text{path } 1} + \left(\frac{\partial L}{\partial w_{ii}} \right)_{\text{path } 2} + \left(\frac{\partial L}{\partial w_{ii}} \right)_{\text{path } 3} + \left(\frac{\partial L}{\partial w_{ii}} \right)_{\text{path } 4} \right]$$



Backward pass (Path-1) for w_{11}^1 :

$$\frac{\partial L}{\partial w_{11}^1} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_1^4} \frac{\partial o_1^4}{\partial net_1^4} \frac{\partial net_1^4}{\partial o_1^3} \frac{\partial o_1^3}{\partial net_1^3} \frac{\partial net_1^3}{\partial o_1^2} \frac{\partial o_1^2}{\partial net_1^2} \frac{\partial net_1^2}{\partial w_{11}^1}$$

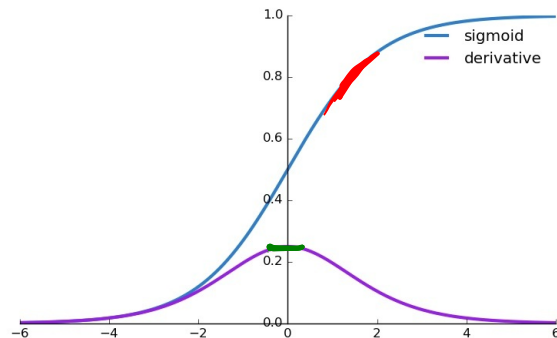
Values for the backward pass:

- $\frac{\partial L}{\partial \hat{y}} = 1$
- $\frac{\partial \hat{y}}{\partial o_1^4} = 0.4$
- $\frac{\partial o_1^4}{\partial net_1^4} = 0.1$
- $\frac{\partial net_1^4}{\partial o_1^3} = 1$
- $\frac{\partial o_1^3}{\partial net_1^3} = 0.5$
- $\frac{\partial net_1^3}{\partial o_1^2} = 1$
- $\frac{\partial o_1^2}{\partial net_1^2} = 0.7$
- $\frac{\partial net_1^2}{\partial w_{11}^1} = 0.1$

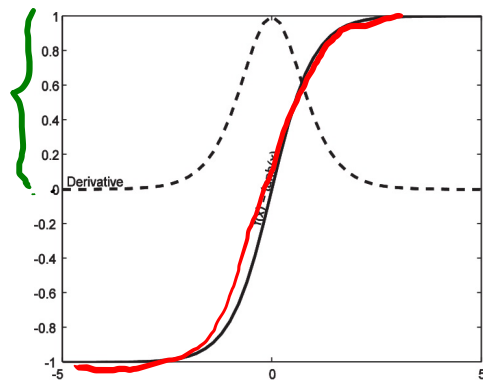
Final result: $\frac{\partial L}{\partial w_{11}^1} = 11 \times 10^{-9}$

Update rule for w_{11}^1 :

$$w_{11}^{1-new} = w_{11}^1 - \eta \times (11 \times 10^{-9}) = 3 - \frac{11 \times 10^{-9}}{9} = ?$$



$$0 < \text{sigmoid}(x) < 1 \rightsquigarrow 0 < \frac{\partial \text{sigmoid}(x)}{\partial x} \leq 0.25$$



$$-1 < \text{tanh}(x) < +1 \rightsquigarrow 0 < \frac{\partial \text{tanh}(x)}{\partial x} < 1$$

α عملاً در شبکه هم با هم می آید ، و زندگی را بر اساس این در می آوریم $Bucle\ pre\ h$

همچنین می توانیم! و عملاً این را هم بر اساس این می بینیم!



Vanishing Gradient.

α

↓ دامین Grad. را در

tanh — Relu

α انتگرال از تابع فعلی:

در این مورد!

$$\text{Relu}(x) \begin{cases} x \\ 0 \end{cases}$$

$$x > 0$$

~ 0

$$x \leq 0$$

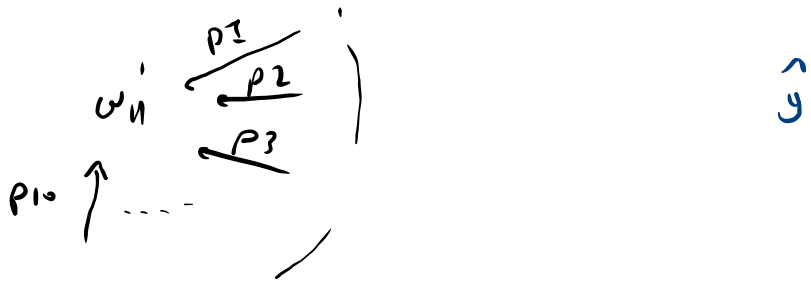
$$\frac{\partial \tanh(x)}{\partial x} < 1$$



$$\frac{\partial \text{Relu}(x)}{\partial x} = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$

Qtd

Relu ~> avoiding underflowing.



tanh: $p_1: 1.13$
 $p_2: 1.13$
 $p_3: 1.13$
 $p_{10}: 1.13$

$$\frac{\partial}{\partial w} \sum p_i = 1.0 \times 1.13 = \underline{1.12}$$

Relu: $p_1: 0$
 $p_2: 1.13$
 $p_3: 1.13$
 $p_{10}: 1.13$

$$\rightarrow \{ p_i = 8 \times 1.13 = \underline{9.04} \}$$

من به دنبال سیس مین میگردم !

stock
predicted

سیس مین بازار مورد

سیس مین تفاوت های کوچک

سیس مین می تواند

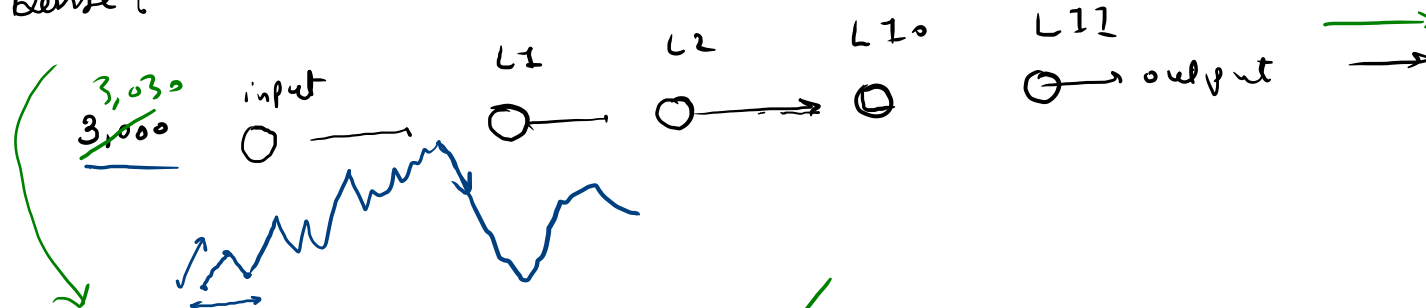
چگونه ؟

۱۴۰۴

[House price forecasting]

آیا شبکه Dense مناسب است؟

Dense :



آیا شبکه Dense می تواند تأثیر سری زمانی در دوره های مختلف را

مدل دوره ها را بیان کند؟

[Sequential Data] / Time Series

price	price
day 1	3,000
day 2	3,030
day 3	3,010
...	...
day 282	2,930

من پر مشتمل ہے کہ بہت زیادہ پیچیدگی کا درجہ حفظ کران

میت زیادہ زیادہ پیچیدگی کا درجہ ہے، و سواں درجہ

نہ ہر Pattern Recognition

↓
نہ ہر حفظ : RNN : نہ ہر ہر : Relament neural net.